Subject : Burgers equation (**Preliminary results**)
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1 Burgers equation

The Burgers equation reads (Debnath (2008)):

$$\boxed{\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} - \nu\frac{\partial^2 u}{\partial x^2} = 0, \quad u > 0} \tag{1}$$

with ν constant in time and space. The Burgers equation is written in conservative form, because we will use a finite volume method to solve this equation, it reads:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} - \frac{\partial}{\partial x}\left(\nu\frac{\partial u}{\partial x}\right) = 0 \tag{2}$$

Applying the finite volume approach:

$$\int_\Omega \frac{\partial u}{\partial t}\, d\Omega + \frac{1}{2}\int_\Omega \frac{\partial u^2}{\partial x}\, d\Omega - \int_\Omega \frac{\partial}{\partial x}\left(\nu\frac{\partial u}{\partial x}\right)\, d\Omega = 0 \tag{3}$$

and after applying Green's theorem it yields:

$$\int_\Omega \frac{\partial u}{\partial t}\, d\Omega + \frac{1}{2}\oint_\Gamma u^2\, d\Gamma - \oint_\Gamma \nu\frac{\partial u}{\partial x}\, d\Gamma = 0 \tag{4}$$

2 Discretization in time and space

The discretization in time and the discretization in space are discussed in separate sections. Due to the applying regularization procedure the viscosity (ν) is increased by an artificial viscosity (Ψ), so the the implementation of $\nu + \Psi$ is a function of x.

2.1 Discretization in time

The used discretization in time reads:

$$\begin{aligned}\int_\Omega \frac{\partial u}{\partial t}\, d\Omega \approx \Delta x \Delta t_{inv}\left(\frac{1}{8}\Delta u_{i-1}^{n+1,p+1} + \frac{6}{8}\Delta u_i^{n+1,p+1} + \frac{1}{8}\Delta u_{i+1}^{n+1,p+1}\right) + \\ + \Delta x \Delta t_{inv}\left(\frac{1}{8}\left(u_{i-1}^{n+1,p} - u_{i-1}^{n}\right) + \frac{6}{8}\left(u_i^{n+1,p} - u_i^{n}\right) + \frac{1}{8}\left(u_{i+1}^{n+1,p} - u_{i+1}^{n}\right)\right)\end{aligned} \tag{5}$$

Special attention is needed if ν is time dependent!

TODO **What to do with time dependent viscosity**

2.2 Discretization in space

Convection-term

$$\frac{1}{2}\oint_{\Gamma} u^2\, d\Gamma = \frac{1}{2}\, u^2\big|_{i+\frac{1}{2}} - \frac{1}{2}\, u^2\big|_{i-\frac{1}{2}} \tag{6}$$

with the approximation $u^2 = (u^{n+\theta,p})^2 + 2u^{n+\theta,p}\theta\Delta u$ the discretization reads:

$$\approx u^{n+\theta,p}_{i+\frac{1}{2}}\theta\Delta u_{i+\frac{1}{2}} - u^{n+\theta,p}_{i-\frac{1}{2}}\theta\Delta u_{i-\frac{1}{2}} + \frac{1}{2}\left(u^{n+\theta,p}_{i+\frac{1}{2}}\right)^2 - \frac{1}{2}\left(u^{n+\theta,p}_{i-\frac{1}{2}}\right)^2 \tag{7}$$

Artificial viscosity term Ψ

Discretization of $\Psi(x,t)$ in time and space needed

Not yet documented

Viscosity term $\nu + \Psi$

With $\varepsilon = \nu + \Psi$ the viscosity term reads:

$$-\oint_{\Gamma} \varepsilon\frac{\partial u}{\partial x}\, d\Gamma = -\left(\left(\varepsilon\frac{\partial u}{\partial x}\right)_{i+\frac{1}{2}} - \left(\varepsilon\frac{\partial u}{\partial x}\right)_{i-\frac{1}{2}}\right) \tag{8}$$

approximated by

$$\Rightarrow -\theta\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x}\Delta u^{n+1,p+1}_{i+1} + \theta\left(\frac{\varepsilon_{i+\frac{1}{2}}}{\Delta x} + \frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x}\right)\Delta u^{n+1,p+1}_{i} - \theta\frac{\varepsilon_{i-\frac{1}{2}}}{\Delta x}\Delta u^{n+1,p+1}_{i-1} + \tag{9}$$

$$-\left\{\left(\varepsilon_{i+\frac{1}{2}}\frac{u^{n+\theta,p}_{i+1} - u^{n+\theta,p}_{i}}{\Delta x} - \varepsilon_{i-\frac{1}{2}}\frac{u^{n+\theta,p}_{i} - c^{n+\theta,p}_{i-1}}{\Delta x}\right)\right\} \tag{10}$$

Artificial viscosity term

Not yet documented

3 Boundary

Because it is a one dimensional equation we have only boundary conditions at the west- and east-side of the domain. We consider only a flow direction in positive x-direction, so $u > 0$.

3.1 West boundary

This is an inflow boundary ($u > 0$).

$$u^{n+1}_{bnd} = f(t) \tag{11}$$

$$u^{n+\theta,p}_{bnd} + \theta\Delta u_{bnd} = f(t) \tag{12}$$

$$\theta\Delta u_{bnd} = f(t) - u^{n+\theta,p}_{bnd} \tag{13}$$

with

$$\Delta u_{bnd} = w_0^{ess} \Delta u_0^{n+1,p+1} + w_1^{ess} \Delta u_1^{n+1,p+1} + w_2^{ess} \Delta u_2^{n+1,p+1}, \tag{14}$$

$$u_{bnd}^{n+\theta,p} = w_0^{ess} u_0^{n+1,p} + w_1^{ess} u_1^{n+1,p} + w_2^{ess} u_2^{n+1,p}, \tag{15}$$

3.2 East boundary

This is an outflow boundary ($u > 0$).

Not yet documented

4 Numerical experiment

In this section we will show the results of the Burgers equation with a Dirichlet boundary conditions at the left side of the domain.

The considered Burgers equation reads:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial u^2}{\partial x} - \frac{\partial}{\partial x}\left((\nu + \Psi)\frac{\partial u}{\partial x}\right) = 0, \qquad u > 0. \tag{16}$$

where

u	Convected velocity, $[\mathrm{m\,s^{-1}}]$.
ν	Constant diffusion coefficient, $[\mathrm{m^2\,s^{-1}}]$.
$\Psi(t)$	Artificial diffusion coefficient, $[\mathrm{m^2\,s^{-1}}]$.

The prescribed boundary condition for the velocity reads

$$u(0,t) = u_{given}(t) \begin{cases} \frac{1}{2}\left(\cos\left(\pi \frac{t_{reg}-t}{t_{reg}}\right) + 1\right) & \text{if } t < t_{reg}, \\ 1 & \text{if } t \geq t_{reg}, \end{cases} \tag{17}$$

where

t_{reg}	The regularization time for the given time-series, [s]
$u_{given}(t)$	Prescribed (given) time-series at the boundary.

5 Experiments without/with artifcial viscosity

In this section results of simulations without and with artificial viscosity will be compared. Several test cases where the viscosity is well resolved on the grid and several cases where is not.

Numerical experiment

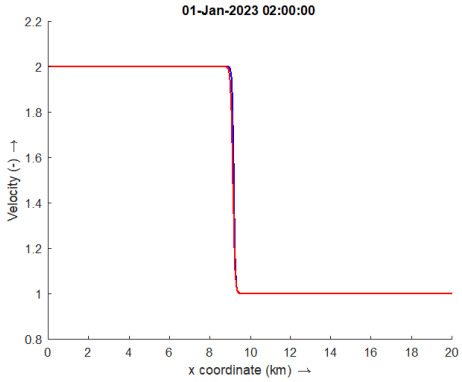
The numerical experiment is performed with the following parameters:

- Length of the domain, $L_x = 20\,000$ [m]
- Grid size, $\Delta x = 10$ [m]
- Start time, $t_{start} = 0$ [s]
- End time, $t_{stop} = 4$ [h]

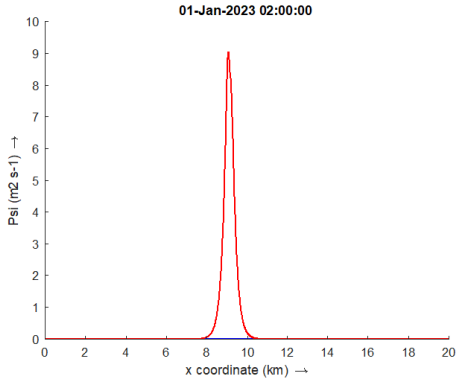
- Timestep, $\Delta t = 5$ [s]
- Regularization time for time-series, $t_{reg} = 1800$ [s]
- Prescribed initial velocity, $u_{initial}(x,t) = 1$ [m s^{-1}]
- Prescribed velocity on boundary at $x = 0$ is given by equation (17), with $u_{given} = 2$ [m s^{-1}].
- No boundary is prescribed at $x = 20\,000$ [m].

The speed of the fully developed shock wave for the inviscid Burgers equation is $u_{shock} = 1.5$ [m s^{-1}] (Debnath, 2008, eq. 8.1.67)

6 Experiment with viscosity of 20 [m s^{-2}]

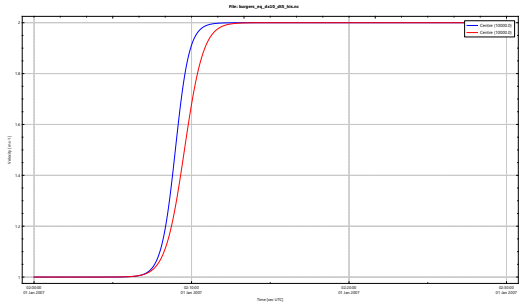


(a) *Velocity along channel at* 2 [h]. *The blue line, without artificial viscosity and the red line with artificial velocity.*

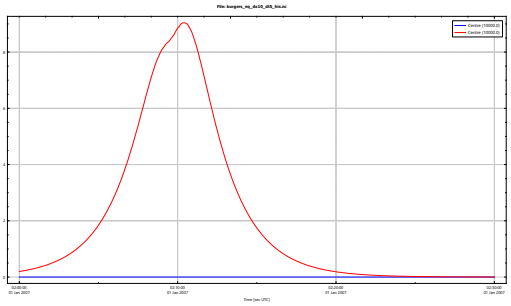


(b) *Used artificial viscoity* Ψ *along channel at* 2 [h]. *The blue line, without artificial viscosity* ($\equiv 0$) *and the red line with artificial velocity.*

Figure 1: *Transect for velocity* $\nu = 20$ [m^2 s^{-1}]

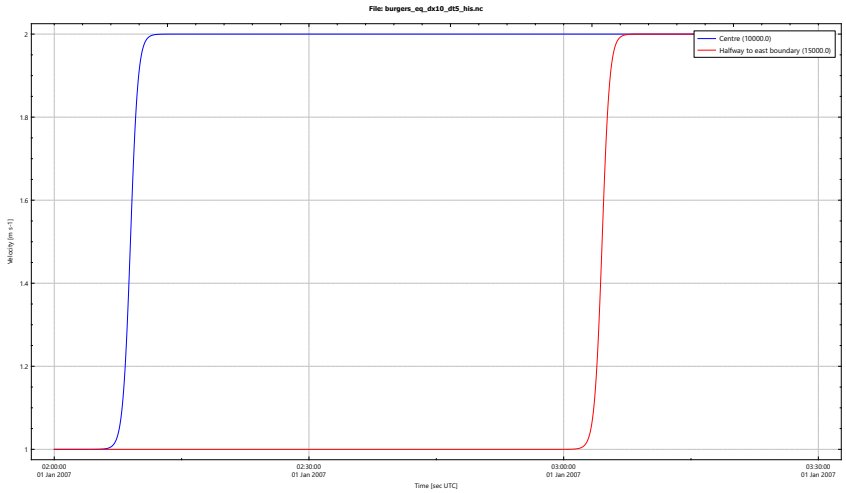


(a) *Time histories of the velocity for the observation point at the centre of the channel. The blue line, without artificial viscosity and the red line with artificial velocity.*

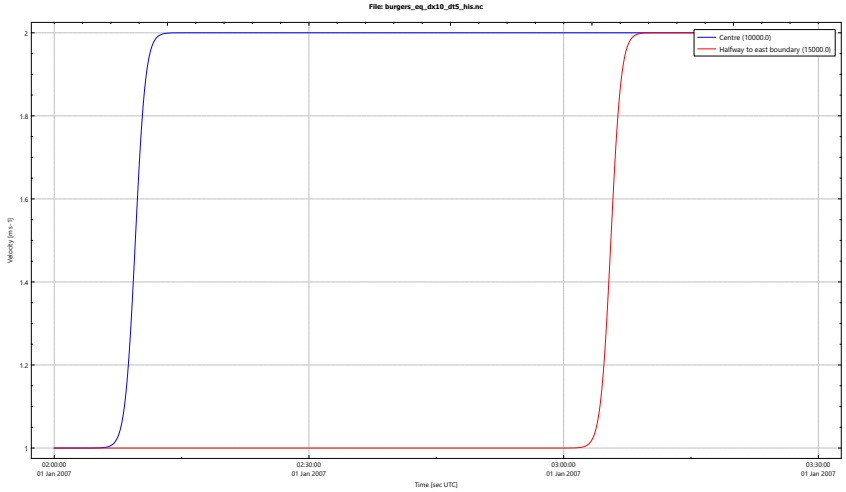


(b) *Time histories of the artificial viscosity* Ψ *for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*

Figure 2: *Time histories for* $\nu = 20$ [m^2 s^{-1}]



(a) *Time histories of the velocity for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*

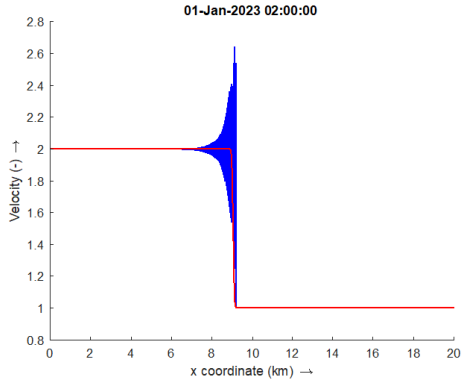


(b) *Time histories of the artificial viscosity* Ψ *for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*

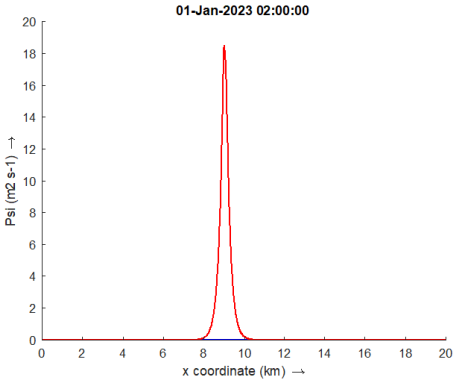
Figure 3: *Time histories for when* $\nu = 20$ [$\mathrm{m^2\,s^{-1}}$]

The wave front has a speed of $u_{shock} = 1.5$ [$\mathrm{m\,s^{-1}}$] for both cases, without and with artificial viscosity (i.e. 1.5000 vs 1.4999 [$\mathrm{m\,s^{-1}}$]).

7 Experiment with viscosity of 0.1 [$\mathrm{m\,s^{-2}}$]

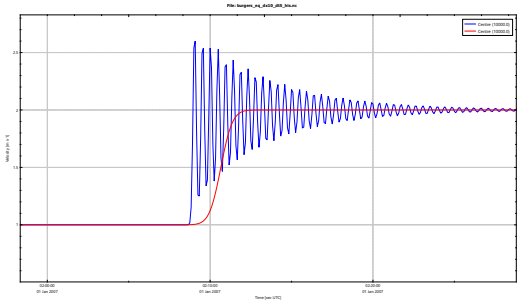


(a) *Velocity along channel at* 2 [h]*. The blue line, without artificial viscosity and the red line with artificial velocity.*

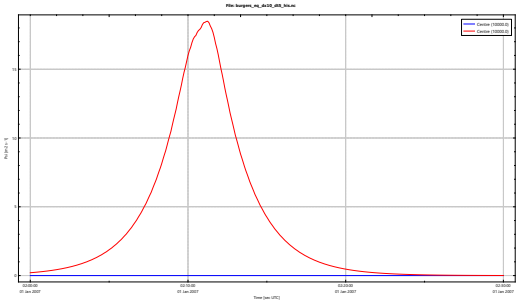


(b) *Used artificial viscoity* Ψ *along channel at* 2 [h]*. The blue line, without artificial viscosity* ($\equiv 0$) *and the red line with artificial velocity.*

Figure 4: *Transect for velocity* $\nu = 0.1$ [$\mathrm{m^2\,s^{-1}}$]

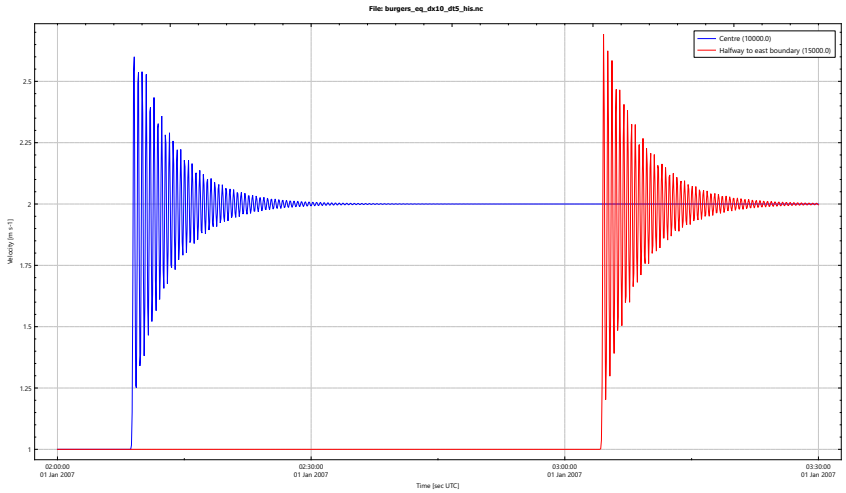


(a) *Time histories of the velocity for the observation point at the centre of the channel. The blue line, without artificial viscosity and the red line with artificial velocity.*

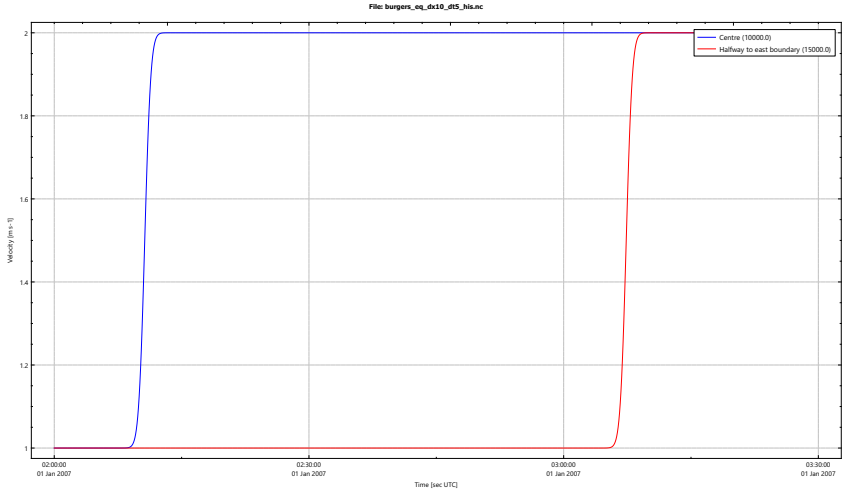


(b) *Time histories of the artificial viscosity* Ψ *for the observation point at the centre of the channel. The blue line, without artificial viscosity* ($\equiv 0$) *and the red line with artificial velocity.*

Figure 5: *Time histories for* $\nu = 0.1$ [$\mathrm{m^2\,s^{-1}}$]



(a) *Time histories of the velocity for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*



(b) *Time histories of the artificial viscosity* Ψ *for the observation point at the* $x = 10\,000$ [m] *(blue line) and at* $x = 15\,000$ [m] *(red line).*

Figure 6: *Time histories for when* $\nu = 0.1$ [$\mathrm{m^2\,s^{-1}}$]

The wave front has a speed of $u_{shock} = 1.5$ [$\mathrm{m\,s^{-1}}$] for both cases, without and with artificial viscosity (i.e. 1.5000 vs 1.5000 [$\mathrm{m\,s^{-1}}$]).

8 Experiment with non linear source term

This experiment is taken from Colombo et al. (2026) and read

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} - \nu\frac{\partial^2 u}{\partial x^2} = u^2, \quad u > 0 \tag{18}$$

With viscosity coefficient $\nu = 0.0$ [$\mathrm{m^2\,s^{-1}}$] and initial condition

$$u(x,0) = \exp(x) + 0.3\exp(-200(x+0.5)^2), \quad x \in [-1,1] \tag{19}$$

and boundary condition at the left side

$$u(-1,t) = \exp(-1). \tag{20}$$

TODO **The expectation is that 2-point Gauss integration is needed.**

Not yet documented

References

Colombo, C., C. Dalmaso, L. O. Müller, and A. Siviglia (2026). “Well-balanced high-order method for non-conservative hyperbolic PDEs with source terms: Application to one-dimensional blood flow equations with gravity”. In: *Journal of Computation Physics* 561. DOI: https://doi.org/10.1016/j.jcp.2026.114975. URL: https://www.sciencedirect.com/science/article/pii/S0021999126003281.

Debnath, Lokenath (2008). *Sir Lighthill and Modern Fluid Dynamics*. Imperial College Press.